

08 – Power Automate Flows (all automation)

Architecture slide lists these automation activities. Here is which flow does what:

Activity from slide	Flow
SharePoint workspace creation	F01
Submission notification	F01
Missing information notification	F02
Funding approval routing	F04
Status change notifications	F02
Reminders & escalations (+ automatic cancellation)	F05
Exception remediation	F08 pattern + Error Log
Completion notifications	F07
Generate PDF and store	F06
Audit trail capture	Dataverse auditing + Status History rows (app & flows) + Notification Log
Document upload from app	F09

General rules for every flow

1. Create it **inside the solution** (Solution → New → Automation → Cloud flow).
2. Use **connection references** and **environment variables** (doc 02).
3. Owner = service account `svc-tvrms`.
4. Name format: `TVRMS - F01 - On Request Submitted`.
5. Use **Try / Catch scopes** (see F08).
6. Choice values in Dataverse are numbers. Write them down once, for example:

Status	Value (example – check yours in the choice editor)
Draft	100000000
Submitted	100000001
Under APPG Review	100000002
More Info Required	100000003
Rejected	100000004
Internal Alignment 1	100000005
Internal Alignment 2	100000006
Pending Funding Approval	100000007
Funding Rejected	100000008
Funding Approved	100000009
In Execution	100000010
On Hold	100000011
Execution Completed	100000012
Closed	100000013
Cancelled	100000014

Common building block: "Send templated email + log it"

Used inside many flows (copy the steps, or make it a child flow **F00 - Send Notification** with inputs: TemplateCode, TVRequestId, ToEmails, ExtraComments).

1. **List rows** (Dataverse) – table *Email Templates*, Filter rows: `tvr_templatecode eq 'REQ_SUBMITTED'` and `tvr_isactive eq true`, Row count 1.

2. **Compose – Subject:**

```
replace(replace(first(outputs('List_template'))['body/value'],
  '{TVID}', triggerOutputs()['body/tvr_tvid']),
  '{Title}', triggerOutputs()['body/tvr_title'])
```

3. **Compose – Body:** same `replace(...)` for `{TVID}`, `{Title}`, `{RequesterName}`, `{Status}`, `{Comments}`, and `{Link}` = `concat(parameters('App URL (tvr_AppUrl)'), '?TVID=', triggerOutputs()['body/tvr_tvid'])`.

4. **Send an email (V2)** (Office 365 Outlook) – To, Subject, Body (HTML). Tip: use **"Send an email from a shared mailbox (V2)"** with `tvrms-noreply@lam.com`.

5. **Add a new row – Notification Logs:** Subject, Template Code, Sent To, Sent On = `utcNow()`, Result = Sent, TV Request = `tvr_tvrequests(<id>)`.

6. If step 4 fails (Configure run after → *has failed*): add Notification Log with Result = Failed + error text.

Sample email (REQ_SUBMITTED)

Subject: [TVRMS] TV-2026-00015 submitted – Plasma etch uniformity test on 300mm wafers

Dear Priya Sharma,

Your Test Vehicle request TV-2026-00015 has been submitted successfully and is now with the APPG team for review.

Request type : New
Required by : 15-Dec-2026
Status : Submitted

Open request: <https://apps.powerapps.com/play/e/.../a/...?TVID=TV-2026-00015>

Regards,
TV Request Management System (automatic email – please do not reply)

F01 – On Request Submitted

Purpose: create SharePoint workspace, send acknowledgement, inform APPG.

Trigger: Dataverse – *When a row is added, modified or deleted*

- Change type: **Modified**
- Table: TV Requests
- Scope: Organization
- Select columns: `tvr_status`
- Filter rows: `tvr_status eq 100000001` (Submitted)

Steps:

1. **Scope – Try**
2. **Initialize variable** FolderList (Array): ["01-Request", "02-Technical Documents", "03-Feasibility & Quotes", "04-Funding Approval", "05-Execution", "06-Final Report & Closure"]
3. **Apply to each** FolderList → SharePoint **Create new folder**
 - Site: env var tvr_SPSiteUrl, List: tvr_SPLibrary
 - Folder path: @{{triggerOutputs()?['body/tvr_tvid']}}/@{{items('Apply_to_each')}}
4. **Update a row** – TV Requests: SharePoint Folder URL = concat(parameters('SharePoint Site URL (tvr_SPSiteUrl)'), '/', parameters('SharePoint Library (tvr_SPLibrary)'), '/', triggerOutputs()?['body/tvr_tvid'])
5. **Get a row by ID** – Users: requester (_tvr_requester_value) → email + full name.
6. (Optional, stricter security) SharePoint **Send an HTTP request** to break inheritance on the TVID folder and grant Edit to requester (see doc 04 §2).
7. Send templated email **REQ_SUBMITTED** → requester.
8. Send templated email **NEW_REQUEST_APPG** → env var tvr_APPGEmail.
9. **Scope – Catch** (run after Try *has failed / timed out*) → F08 pattern.

Test with sample: submit TV-2026-00015 → folder TV Requests/TV-2026-00015/... with 6 sub-folders exists, Priya gets email, APPG mailbox gets email, 2 Notification Log rows.

F02 – Status Change Notifier

Purpose: one flow that sends the right email for every status change.

Trigger: Dataverse *When a row is modified* – TV Requests – Select columns tvr_status (no filter).

Steps:

1. Get requester, APPG owner, and cost center owner emails (Get a row by ID ×3; use *Configure run after* to continue if a lookup is empty).
2. Get the latest comment (List rows *Request Comments*, filter on this request, order by createdon desc, top 1) → used as {Comments}.
3. **Switch** on triggerOutputs()?['body/tvr_status']:

Case (status)	Template	To	CC
Under APPG Review	REVIEW_STARTED	Requester	–
More Info Required	MORE_INFO	Requester	APPG owner
Rejected	REJECTED	Requester	APPG team
Internal Alignment 1 / 2	ALIGNMENT_UPDATE	Requester, stakeholders	APPG owner
Funding Approved	FUNDING_APPROVED	Requester, APPG owner	CCO
Funding Rejected	FUNDING_REJECTED	Requester, APPG owner	CCO
In Execution / On Hold	STATUS_UPDATE	Requester, stakeholders	–
Execution Completed	STATUS_UPDATE	Requester	–

Case (status)	Template	To	CC
Cancelled	CANCELLED	Requester	APPG owner
Submitted, Pending Funding Approval, Closed	<i>(do nothing – F01/F04/F07 handle these)</i>		

4. Send templated email + log (common block).

Exception notification: a second small trigger flow **F02b** on *Risk Issues* "When a row is added" with filter `tvr_severity eq <High>` → email APPG owner + requester with template EXCEPTION_RAISED.

F04 – Funding Approval Routing

Trigger: TV Requests modified, Select columns `tvr_status`, Filter `tvr_status eq 100000007` (Pending Funding Approval).

Steps:

1. **Get a row by ID** – Cost Centers (`_tvr_costcenter_value`) → Owner, Owner's Manager.
2. **Get a row by ID** – Users → CCO email, Manager email.
3. **Get a row by ID** – Feasibility Options (selected option) → quoted cost, institution.
4. **List rows** – App Settings, filter `tvr_name eq 'HighValueFundingLimit'` → limit (e.g. 50000).
5. **Perform an unbound action** `AddUserToRecordTeam` → add CCO to the request's access team (so CCO can open the record) – doc 05 §2.
6. **Add a new row** – Funding Approvals: Request, Cost Center, Approver = CCO, Level 1, Requested Amount = quoted cost, Decision = Pending, Requested On = `utcNow()`.
7. **Start and wait for an approval** (Approvals connector)
 - Type: **Approve/Reject – First to respond**
 - Title: `Funding approval: @{TVID} – USD @formatNumber(QuotedCost, 'N0')`
 - Assigned to: CCO email
 - Details (markdown):

```

**Request:** TV-2026-00015 – Plasma etch uniformity test
**Requester:** Priya Sharma (Process Engineering)
**Cost center:** CC-4100 Etch R&D
**Selected option:** Stanford Nanofab Lab – USD 39,500 – 10 weeks
**Justification:** Customer X reports 4% non-uniformity...

```

- Item link: app deep link `?TVID=...`

8. **Update a row** – Funding Approval: Decision = outcome, Comments = `first(body('Start_and_wait_for_an_approval')?['responses'])?['comments']`, Decided On = `utcNow()`.
9. **Condition:** Outcome = "Approve" **and** amount > limit?
 - Yes → repeat steps 6–8 with **Level 2**, Approver = Owner's Manager.
10. **Condition:** final outcome = Approve?
 - **Yes:** Update TV Request → Status = Funding Approved, Approved Amount = quoted cost, Status Changed On = `utcNow()`; add Status History row (Changed By = approver).
 - **No:** Update TV Request → Status = Funding Rejected, Reason = approver comments; add Status History row.
11. F02 then sends FUNDING_APPROVED / FUNDING_REJECTED automatically.

Note: a flow run can wait at most **30 days**. F05 sends reminders to the CCO while the approval is pending. If 30 days pass, the run ends; F05 flags it to the APPG owner.

Sample: quoted 39,500 ≤ 50,000 → only Maria Lopez approves → status Funding Approved. Quoted 62,000 → Maria approves, then James Wilson approves → Funding Approved.

F05 – Daily Reminders, Escalations and Auto-Cancellation

Trigger: Recurrence – every 1 day at 08:00, time zone = LAM local, on weekdays.

Steps:

1. Read settings: ReminderAfterDays (3), EscalationAfterDays (5), AutoCancelAfterDays (30).

2. **List rows** – TV Requests waiting for an action:

```
(tvr_status eq 100000001 or tvr_status eq 100000002 or tvr_status eq 100000003
or tvr_status eq 100000005 or tvr_status eq 100000006 or tvr_status eq 100000007)
and tvr_statuschangedon le @{addDays(utcNow(), mul(-1, int(variables('ReminderDays'))))}
```

3. **Apply to each** request:

- Compose **DaysWaiting** = `div(sub(ticks(utcNow()), ticks(items('Apply_to_each')?['tvr_statuschangedon'])), 864000000000)`
- Decide **who must act**: More Info Required → requester; Pending Funding → CCO; other statuses → APPG owner (or APPG team if no owner).
- **Condition A – auto-cancel**: status = More Info Required **and** DaysWaiting ≥ 30 → Update row: Status = Cancelled, Reason = "Auto-cancelled: no response for 30 days"; add Status History (Changed By = system). (F02 sends the CANCELLED email.)
- **Condition B – escalate**: DaysWaiting ≥ 5 → Office 365 Users **Get manager (V2)** of the person who must act → send ESCALATION email to manager, CC the person.
- **Else – reminder**: send REMINDER email to the person who must act.

4. **List rows** – Milestones where `tvr_plannedend lt @{utcNow()}` and status ne Completed → reminder to APPG owner ("Milestone overdue").

5. **List rows** – TV Requests In Execution where Expected Completion < today → email APPG owner.

Sample run (12-Oct-2026):

TVID	Status	Days waiting	Action
TV-2026-00018	Submitted	3	Reminder to APPG team
TV-2026-00016	More Info Required	6	Escalation to Priya's manager
TV-2026-00009	More Info Required	31	Auto-cancelled
TV-2026-00012	Pending Funding Approval	4	Reminder to CCO

Working days only? (ASSUMPTION: calendar days). If the business wants working days, keep a *Holiday* table and count only Mon–Fri non-holiday days.

F06 – Generate PDF and Store

Make two flows:

- **F06a – Build Request PDF (child flow)** – trigger *Manually trigger a flow* with input `TVID`.
- **F06 – Export PDF (from app)** – trigger *Power Apps (V2)* with input `TVID` → **Run a Child Flow F06a** → **Respond to a PowerApp or flow** with `fileurl`.

F06a steps:

1. List rows TV Requests `tvr_tvid eq '{@triggerBody()}['text']'` (expand requester, cost center).
2. List rows Feasibility Options, Funding Approvals, Milestones for this request.
3. **Create HTML table** for options / approvals / milestones.
4. **Compose** full HTML:

```
<html><body style="font-family:Segoe UI">
<h1>TV Request Summary – TV-2026-00015</h1>
<p><b>Title:</b> Plasma etch uniformity test ... <b>Status:</b> Closed</p>
<h2>Feasibility Options</h2> @body('Create_HTML_table_options')
<h2>Funding Approvals</h2> @body('Create_HTML_table_approvals')
<h2>Milestones</h2> @body('Create_HTML_table_milestones')
<h2>Closure</h2><p>Outcome: ... Lessons learned: ...</p>
<p>Generated on @formatDateTime(utcNow(),'dd-MMM-yyyy HH:mm') UTC</p>
</body></html>
```

5. OneDrive for Business **Create file** `/TVRMS-Temp/{TVID}.html` with the HTML.
6. OneDrive **Convert file** (by Id) → Target type **PDF**.
7. SharePoint **Create file** → folder `/@{TVID}/06-Final Report & Closure` (or `01-Request` if not closed), name `@{TVID}_Summary_@formatDateTime(utcNow(),'yyyyMMdd').pdf`.
8. Add a Request Document row (Type = Approval Evidence / Final Report).
9. OneDrive **Delete file** (temp HTML).
10. **Respond to a PowerApp or flow:** `fileurl` = SharePoint file link.

(Option: if LAM has a Word template, use *Populate a Microsoft Word template* (premium) instead of HTML.)

F07 – On Request Closed (Completion Notification)

Trigger: TV Requests modified, Select `tvr_status`, Filter `tvr_status eq 100000013` (Closed).

Steps:

1. **Run a Child Flow** F06a → closure summary PDF saved in `06-Final Report & Closure`.
2. Find the final report link (List rows Request Documents, Type = Final Report, top 1).
3. Send templated email **COMPLETED** → requester, APPG owner, CCO, stakeholders, with links to the final report and the summary PDF ("Report Available Notification").
4. If Request Type = Recurring → send **NEXT_OCCURRENCE** reminder to requester: "Your next Quarterly TV request is due. Use 'Repeat' to create it."

F08 – Error handling pattern (Exception Remediation)

Put every flow's steps inside **Scope – Try**. Then add **Scope – Catch** with *Configure run after* = **has failed, has timed out**:

1. Add a new row – Error Logs:

- Flow Name: `@{workflow()?['tags']?['flowDisplayName']}`
- Error: `@{string(result('Try'))}`
- Run URL:

```
concat('https://make.powerautomate.com/environments/', workflow()?['tags']?['environmentName'],
      '/flows/', workflow()?['name'], '/runs/', workflow()?['run']?['name'])
```

- TV Request: the trigger row id.

2. Send email to env var `tvr_AdminEmail`: "TVRMS flow failed – {Flow} – {TVID}" + Run URL.

3. **Terminate** – Status: Failed (so the run shows red in history).

Admin opens **Monitoring → Open Errors** in the model-driven app, fixes the cause, and **resubmits** the failed run from flow run history. Then sets Error Log Status = Fixed.

Also set **Retry policy** (Settings of each action) for SharePoint/Outlook actions: *Exponential, 4 retries*.

F09 – Upload Document (called from the app)

Trigger: Power Apps (V2) with inputs:

Input	Type	Sample
TVID	Text	TV-2026-00015
DocType	Text	Technical Specification
Folder	Text	02-Technical Documents
file	File	Etch_Test_Spec_v1.pdf

Steps:

1. SharePoint **Create file**

- Folder path: `/@{parameters('SharePoint Library (tvr_SPLibrary)')}/@{triggerBody()['text']}/@{triggerBody()['text_2']}`
- File name: `@{triggerBody()?['file']?['name']}`
- File content: `@{triggerBody()?['file']?['contentBytes']}` (If the folder does not exist yet, SharePoint creates it. Same file name again = new version.)

2. SharePoint **Update file properties** (Id = `outputs('Create_file')?['body/ItemId']`): TVID, DocumentType, Phase.

3. **List rows** TV Requests where `tvr_tvid eq '@{triggerBody()['text']}'`.

4. **Add a new row** – Request Documents: File Name, Document Type, Folder, SharePoint URL, TV Request.

5. **Respond to a PowerApp or flow** – `fileurl`.

App call (from doc 06):

```
'F09-UploadDocument'.Run(varNew.TVID, "Technical Specification", "02-Technical Documents",
  { file: { name: f.Name, contentBytes: f.Value } })
```

F10 – Add Stakeholder (optional, from app)

Power Apps (V2) trigger: TVID, StakeholderEmail → Get user → `AddUserToRecordTeam` (template "TVR Request Team") → email STAKEHOLDER_ADDED.

Flow testing checklist

Flow	How to test	Pass when
F01	Submit a request	6 folders created, 2 emails, URL saved
F02	Change status to More Info Required	Requester gets MORE_INFO email with comment
F04	Send for funding (amount 39,500)	CCO gets approval in Outlook/Teams; decision saved
F04	Amount 62,000	Two approvals in sequence
F05	Set Status Changed On to 31 days ago on a More Info request; run flow manually	Request auto-cancelled
F06	Click Export PDF	PDF opens; file in SharePoint
F07	Close request	Completion email with links
F08	Temporarily rename SharePoint library in env var; submit	Error Log row + admin email
F09	Upload 2 files	Both in SharePoint + 2 Request Document rows